

Assignment 3

Neuromembrane Lab 3

The purpose of this assignment is to help you understand how membrane permeability and ion concentrations contribute to the resting membrane potential. We will go through multiple simulations. **Be sure to answer all the questions throughout this assignment.** The questions are in red.

Goals:

1. To understand equilibrium potentials by experimenting with ion concentrations and calculating them using the Nernst equation
2. To understand how the resting potential depends on the relative permeability (conductances) of Na^+ and K^+ .
3. To understand how signals can be generated by changing the ratio of the conductances to Na (g_{Na}) and K (g_{K}).
4. To experiment with neurotoxins and observe their effects on the action potential.

Background:

Understanding equilibrium potentials is fundamental to understanding cell signaling and excitability.

In this tutorial you can experiment with intracellular and extracellular Na and K concentrations to see how they determine E_{Na} and E_{K} , the equilibrium potentials for these ions. You can then vary the ratio between the cell's conductance to K and to Na to understand how ion selectivity determines the membrane potential (V_m). In subsequent tutorials you will see that V_m controls the opening of ion channels and the generation of signals in neurons, skeletal muscle, and heart muscle. Thus, excitability depends ultimately on precisely regulated ionic concentrations.

Equilibrium potentials are calculated, not measured.

While V_m is a cell property that is observed and measured, E_{ion} is calculated by the Nernst equation. The equilibrium potential for an ion (E_{ion}) and the membrane potential (V_m) are equal only under one special circumstance: when the cell is permeable (has conductance) only to that ion. E_{ion} depends only on three variables: (1) the ratio of the concentrations of that ion inside and outside of the cell, (2) the absolute temperature (273 degrees + experimental temperature in centigrade), and (3) the valence of the ion.

If the ion concentrations do not change, E_{Na} and E_{K} do not change.

V_m can soar from negative to positive values and back again but the equilibrium potentials remain steady at their values. If the cell becomes permeable only to Na ions, then V_m will soar to E_{Na} . If it then becomes permeable only to K ions, V_m will plunge back to E_{K} . In fact, when Na and K are the only ions to consider, V_m is trapped between E_{Na} and E_{K} . Notice how the changes in V_m , changes that ultimately constitute signals, can be generated simply by changing the relative ionic conductances (permeabilities) of the cell with time.

Conversely, if the extracellular ion concentrations change, then E_{K} or E_{Na} will change with potentially harmful consequences.

A change in an equilibrium potential can thus greatly affect the V_m of cells and lead to pathological conditions. For example, a slight increase in the $[\text{K}]$ in the blood due to dehydration can have very serious consequences.

Action Potentials using Neuromembrane Simulator

<https://neuromembrane.ualberta.ca/>

How to use the program:

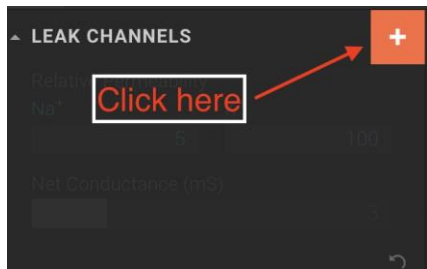
1. Select “Log in as a guest” button.
2. You’ll now be inside of the action potential simulation.
 - a. **Tip:** Double-check that this is the simulation you’re in by looking the at the top of the panel to the left. It should say “Action Potential Simulation.”
3. **Main page sections:**
 - **Left Section:** Allows you to adjust the parameters necessary for the simulation and is composed of several boxes that allow you to control the leak channels, Na⁺/K⁺ pump, initial membrane voltage, ion concentration settings, etc.
 - **Middle Section:** Shows a plain lipid bilayer upon which you can build your excitable membrane. Clicking the “+” sign at the top right of the box adds that particular item to the membrane in the right half of the window and allows you to adjust more detailed parameters for that item. These parameters can be adjusted by either clicking on the number, directly typing it in, or by dragging the slider left and right. The small counterclockwise arrow in the bottom right hand corner of each box allows you to revert back to default values.
 - **Right Section:** Shows the membrane simulation. The Na⁺ and K⁺ ions can be seen in the top half of the window, which represents the intracellular side. Adjusting the concentrations of the ions results in a greater (or lower) number of ions in the appropriate field of view.

Experiment 1 – Recording from a glial cell

Background: Many glial cells are permeable only to K ions; they have no measurable Na permeability. Because of this, the glial Vm acts as a "K electrode": That is, its resting membrane potential will always be equal to its calculated EK. Long ago Steve Kuffler realized this and used the Vm of glial cells to report the value of the [K] in the extracellular spaces of the leech CNS when neurons fired trains of action potentials (Kuffler and Nicholls 1966).

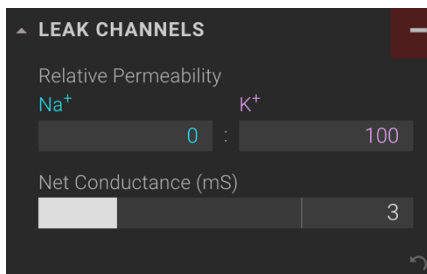
Steps to complete Experiment 1:

1. Let's add ion channels and the sodium-potassium pump! Within the control panel on the left-hand side, you'll see several components that you can add to your cell membrane. These components include leak channels, Na⁺ channel, K⁺ channel, and the Na⁺/K⁺ pump. ***Add all these components by clicking the + button.*** See screenshot below.



Tip: Once these elements have been added to the membrane, you can hover your mouse over them to see their labels.

2. Now that these elements are reflected on your cell membrane (see screenshot below), let's adjust the relative permeability of the membrane to reflect that of the glial cell. Under "Leak Channels", change the relatively permeability of Na⁺ to "0". Keep K⁺ at "100". See screenshot below. The cell is now only permeable to potassium ions.



3. During this experiment, we will **NOT** be applying an “external positive current”. We are most interested in understanding how changing the permeability/conductance of Na^+ and K^+ impacts the resting membrane potential and equilibrium potentials. As such, change your external current panel so that we are **NOT** administering any stimulus (all current should be 0). It should now look like this:

Current (μA)	Time (ms)
0	10
0	2
0	10
0	2
0	16

4. Now, take note of a few numbers on your screen:

- a. What is the current resting membrane potential? (This can be found under “Membrane Settings”, called GHK Membrane Potential)

Resting membrane potential = -95.0 mV

- b. And what is the equilibrium, or reversal potential of K^+ ? (This can be found under “ K^+ Channel”, called Reversal Potential)

Equilibrium potential of K^+ (E_K) = -95.0 mV

5. Now let’s see this neuron in action! Select the “Create Simulation” button located under the cell membrane. Select the play button at the bottom of the cell membrane to begin the animation. Take note of the movement of ions.

- a. **Note:** You should NOT have gotten an action potential, given we did not administer any stimulus.

6. Record the voltage using the “Voltage vs Time” graph. To do so, you will place your cursor over the highest point the line reached and write down the first value. You can also click “summary” to see the left panel expand and provide you more information, including the voltage.

Membrane Voltage (V_m)= -95.0 mV

7. Click the button to the left of the play button to return to the settings page. We will now be manipulating the concentration of K^+ **outside** of the glial cell (in the extracellular space). Typically, the normal concentration of external K^+ is 4 mM. This is the value that the simulation

automatically starts at. You will run a total of FIVE more simulations, only changing the external K^+ concentration each time. To change $[K^+]_o$, you will go to “concentration settings” and adjust the $[K^+]$ Outside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right. **For this table, use GHK Membrane Potential as your values for V_m .**

Track your results in the chart below:

Simulation #	$[K^+]_i$	$[K^+]_o$	E_k	V_m
1	140	4	-95.0 mV	-95.0 mV
2	140	8	-76.4 mV	-76.4 mV
3	140	10	-70.5 mV	-70.5 mV
4	140	20	-52.0 mV	-52.0 mV
5	140	30	-41.1 mV	-41.1 mV
6	140	40	-33.5 mV	-33.5 mV

8. Experiment summary explanation: As potassium outside of the glial cell increases $[K^+]_o$, what happens to the equilibrium potential of potassium (E_k)? What about the membrane voltage (V_m)? Are these values the same or different? EXPLAIN WHY.

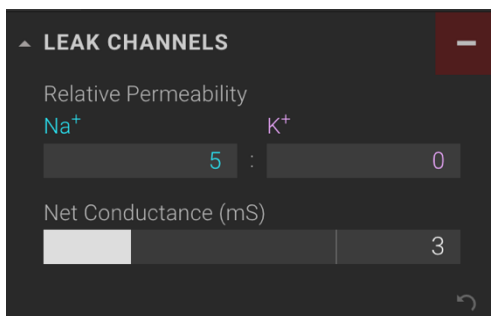
As you increase potassium outside of the glial cell, both the equilibrium potential of potassium AND the membrane voltage increase (at the same rate). This is because the glial cell is only permeable to potassium, so the membrane potential will always equal the equilibrium potential of potassium.

Experiment 2 – Recording from a Na^+ -permeable membrane

Background: Unlike the experiment above, there is no real-world scenario in which a cell is only permeable to Na^+ . As such, in this experiment, we will be doing something that can only be manipulated virtually, and not in real life. We will be creating a cell that is only permeable to Na^+ .

Steps to complete Experiment 2:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. Opposite to the experiment above, let’s adjust the relative permeability of the membrane to now reflect that of a cell that is only permeable to Na^+ . Under “Leak Channels”, change the relatively permeability of K^+ to “0”. Keep Na^+ at “5”. See screenshot below.



4. Similar to the experiment above, we will **NOT** be applying an “external positive current”. We are most interested in understanding how changing the permeability/conductance of Na^+ and K^+ impacts the resting membrane potential and equilibrium potentials. As such, change your external current panel so that we are **NOT** administering any stimulus (all current should be 0). It should now look like this:

Current (μA)	Time (ms)
0	10
0	2
0	10
0	2
0	16

5. Now, take note of a few numbers on your screen:

- a. What is the current resting membrane potential? (This can be found under “Membrane Settings”, called GHK Membrane Potential)

Resting membrane potential = 70.5 mV

- b. And what is the equilibrium, or reversal potential of Na^+ ? (This can be found under “ Na^+ Channel”, called Reversal Potential)

Equilibrium potential of Na^+ (E_{Na}) = 70.5 mV

6. Now let's see this neuron in action! Select the “Create Simulation” button located under the cell membrane. Select the play button at the bottom of the cell membrane to begin the animation. Take note of the movement of ions.

- a. **Note:** You should NOT have gotten an action potential, given we did not administer any stimulus.

7. Record the voltage using the “Voltage vs Time” graph. To do so, you will place your cursor over the highest point the line reached and write down the first value. You can also click “summary” to see the left panel expand and provide you more information, including the voltage.

Membrane Voltage (V_m) = 70.5 mV

8. Click the button to the left of the play button to return to the settings page. We will now be manipulating the concentration of Na^+ **outside** of the cell (in the extracellular space). Typically, the normal concentration of external Na^+ is 140 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE more simulations, only changing the external Na^+ concentration each time. To change $[\text{Na}^+]_o$, you will go to “concentration settings” and adjust the $[\text{Na}^+]$ Outside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right. **For this table, use GHK Membrane Potential as your values for V_m .** Track your results in the chart below:

Simulation #	$[\text{Na}^+]_i$	$[\text{Na}^+]_o$	E_{Na}	V_m
1	10	80	55.5 mV	55.5 mV
2	10	100	61.5 mV	61.5 mV
3	10	120	66.4 mV	66.4 mV
4	10	140	70.5 mV	70.5 mV
5	10	180	77.2 mV	77.2 mV
6	10	200	80.0 mV	80.0 mV

9. Experiment summary explanation: As sodium outside of the cell increases $[\text{Na}^+]_o$, what happens to the equilibrium potential of sodium (E_{Na})? What about the membrane voltage (V_m)? Are these values the same or different? EXPLAIN WHY.

As sodium outside of the cell increases $[\text{Na}^+]_o$, both the equilibrium potential of sodium (E_{Na}) AND the membrane voltage (V_m) increase (at the same rate). There is an increase because there is a stronger theoretical drive for Na^+ to rush down its “chemical gradient” into the cell where its concentration is lower (this is the concept of “electrostatic pressure”).

10. If you run the simulation for this experiment, you will notice that V_m does not equal E_{Na} as evidenced by your answers to questions 5b and 7. A similar trend happens for experiment 1 at around $[\text{K}^+]_o = 10$. Explain why the calculated value (GHK membrane potential) of the Nernst equation eventually does not equal the actual value of V_m . **Hint:** The action of one of the components of the neuron membrane is involved.

They are actually the same every time in experiments #1 and #2 (questions 5b and 7).

Experiment 3 – What determines the "resting potential"?

Background: A typical neuron is permeable to both K and Na ions, although far less so to Na than to K. What, then, determines the value of V_m at which the neuron "rests," V_{rest} ? (In this tutorial we have simplified the resting potential, whose value may be influenced by factors in addition to Na and K permeability. For example, the resting potential may have a contribution from "electrogenic" ion pumps that transport charges unequally across the membrane.)

Steps to complete Experiment 3:

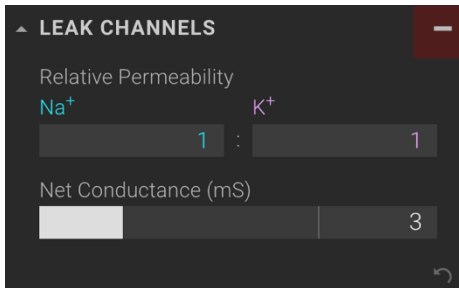
1. Log out of the neuromembrane simulation and log back in as a "guest". This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the "+" next to each component.
3. Similar to the experiments above, we will **NOT** be applying an "external positive current". We are most interested in understanding how changing the permeability/conductance of Na^+ and K^+ impacts the resting membrane potential and equilibrium potentials. As such, change your external current panel so that we are **NOT** administering any stimulus (all current should be 0). It should now look like this:

Current (μA)	Time (ms)
0	10
0	2
0	10
0	2
0	16

4. **STOP!** Before moving on, take note of the resting membrane potential. (This can be found under "Membrane Settings", called GHK Membrane Potential)

Resting membrane potential = -68.0 mV

5. We will now be manipulating the permeability ratio of the leak channels from 1:1 (i.e., an equal ratio) all the way up to 1:20 (i.e., the membrane will be 20 times more permeable to K^+ than Na^+). Typically, the normal permeability ratio is 5:100 (same as 1:20, which is the reduced ratio). This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the relative permeability of Na^+ to K^+ each time. To begin, under “Leak Channels”, change the relatively permeability of Na^+ to “1” and K^+ to “1”. See screenshot below.



6. **STOP!** Take note of the resting membrane potential now. (This can be found under “Membrane Settings”, called GHK Membrane Potential)

Resting membrane potential = **-1.10 mV**

7. Use the values listed below for the rest of your **FIVE** simulations. We will keep the permeability of Na^+ at “1” but increase the permeability of K^+ by “5 steps” each time. Record the resting membrane potential each time.

Simulation #	Permeability of Na^+	Permeability of K^+	RMP
1	1	1	-1.10 mV
2	1	5	-39.8 mV
3	1	10	-55.0 mV
4	1	15	-62.9 mV
5	1	20	-68.0 mV

8. Experiment summary explanation: How does manipulating the permeability ratio of Na^+ to K^+ impact the resting membrane potential? In other words, as the cell becomes more permeable to potassium (as compared to sodium), how does the membrane voltage change? Does it get closer to the reversal potential of potassium (-95) or sodium (70.5)? EXPLAIN.

Manipulating the permeability ratio of Na^+ to K^+ by increasing K^+ results in a decrease of the resting membrane potential (becomes more negative). In other words, as the cell becomes

increasingly permeable to potassium (while sodium remains constant), this lowers the membrane voltage closer to the reversal potential of potassium (-95mV) rather than that of sodium (70.5mV). The reason for this is because potassium is more permeable, it has a greater influence on the membrane potential (while sodium has minimal effect remaining constant at "1"). As the GHK equation describes, the ion with the highest permeability has the greatest influence on membrane potential.

Experiment 4 – How does the resting potential depend on ion concentrations?

Background: We now want to see how the typical relative permeability ratio (1:20 or 5:100) impacts how changes in ion concentrations will affect the membrane voltage. We will first be changing $[K^+]_o$.

Steps to complete Experiment 4:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. As you will see, the relative permeability ratio of Na^+ to K^+ has returned to 5:100 (remember from the experiment above, this is the same fraction as 1:20, meaning the membrane is 20 times more permeable to potassium than sodium). **KEEP IT AT 5 to 100.** Note that the resting membrane potential is currently **-68.0 mV**.
4. Similar to the experiments above, we will **NOT** be applying an “external positive current”. As such, change your external current panel so that we are **NOT** administering any stimulus (all current should be 0). It should now look like this:

Current (μA)	Time (ms)
0	10
0	2
0	10
0	2
0	16

5. We will now be manipulating the concentration of **K^+ outside** of the cell (in the extracellular space). Typically, the normal concentration of external K^+ is 4 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the external K^+ concentration each time. To change $[K^+]_o$, you will go to “concentration settings” and

adjust the $[K^+]_{\text{Outside}}$ (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right.

6. Use the values listed below for your FIVE simulations. We will keep increasing the amount of external K^+ each time. Record the resting membrane potential each time.

Note: Permeability ratio should be 5:100 or 1:20

Simulation #	$[K^+]_o$	RMP
1	4	-68.0 mV
2	20	-44.1 mV
3	50	-24.1 mV
4	80	-12.8 mV
5	100	-7.3 mV

7. Experiment summary explanation: When the membrane at rest is 20 times more permeable to potassium, how does manipulating the concentration of **K^+ outside** of the cell impact the resting membrane potential? EXPLAIN WHY.

When the membrane at rest is 20 times more permeable to potassium, increasing the concentration of K^+ outside of the cell increases the resting membrane potential (becomes more positive). This is because there is a reduced chemical gradient for K^+ leaving the cell as there is already a higher concentration outside (reduced need to leave) leading to more positive ions inside the cell and therefore an increased membrane potential.

Experiment 5 – How does the resting potential depend on ion concentrations?

Background: We now want to see how the typical relative permeability ratio (1:20 or 5:100) impacts how changes in ion concentrations will affect the membrane voltage. Now we will be changing $[\text{Na}^+]_o$.

Steps to complete Experiment 5:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. As you will see, the relative permeability ratio of Na^+ to K^+ has returned to 5:100 (remember from the experiment above, this is the same fraction as 1:20, meaning the membrane is 20 times more permeable to potassium than sodium). **KEEP IT AT 5 to 100.** Note that the resting membrane potential is currently **-68.0 mV**.
4. Similar to the experiments above, we will **NOT** be applying an “external positive current”. As such, change your external current panel so that we are **NOT** administering any stimulus (all current should be 0). It should now look like this:

Current (μA)	Time (ms)
0	10
0	2
0	10
0	2
0	16

5. We will now be manipulating the concentration of **Na⁺ outside** of the cell (in the extracellular space). Typically, the normal concentration of external Na⁺ is 140 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the external Na⁺ concentration each time. To change [Na⁺]_o, you will go to “concentration settings” and adjust the [Na⁺] Outside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right.

6. Use the values listed below for your FIVE simulations. We will keep increasing the amount of external Na⁺ each time. Record the resting membrane potential each time.

Note: Permeability ratio should be 5:100 or 1:20

Simulation #	[Na ⁺] _o	RMP
1	100	-73.4 mV
2	120	-70.6 mV
3	140	-68.0 mV
4	160	-65.7 mV
5	200	-61.6 mV

7. Experiment summary explanation: When the membrane at rest is 20 times more permeable to potassium, how does manipulating the concentration of **Na⁺ outside** of the cell impact the resting membrane potential? Did changing **K⁺ outside** or **Na⁺ outside** concentration outside of the cell alter RMP more? Use Experiment #4 to compare. EXPLAIN WHY.

When the membrane at rest is 20 times more permeable to potassium, manipulating the concentration of Na⁺ outside of the cell results in an increase of the resting membrane potential (becomes more positive). This is a similar result as increasing the K⁺ outside the cell in experiment four, however the magnitude of increase in resting membrane potential here is smaller than it is for potassium (again, due to the fact that potassium is 20 times more permeable and therefore has greater impact).

Experiment 6 – Effects of Tetrodotoxin (TTX) on Action Potential Firing

Background: At the conclusion of Ian Fleming's book "From Russia, With Love," we see agent 007, James Bond, crumple to the floor after a miniscule wound from a pinprick, delivered, of course, by a beautiful woman. In the sequel, "Dr. No," Fleming reveals that the pinpoint was poisoned with TTX from the fugu fish. Prompt application of artificial respiration had saved Bond's life. **TTX is a highly specific blocker of Na channels.**

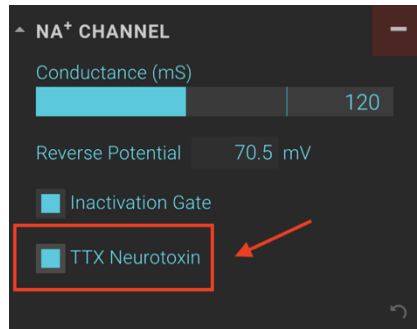
Steps to complete Experiment 6:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na⁺ channels, K⁺ channels, and the Na⁺/K⁺ pump to the membrane by clicking the “+” next to each component.
3. Unlike the experiments above, we want to inject **ONE** stimulus current. Additionally, this current needs to be strong enough for the action potential to reach threshold. As such, change your external current panel to the following:

Current (μA)	Time (ms)
0	10
25	2
0	10
0	2

0	16
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- Next, select the “TTX Neurotoxin” under the “Na⁺ Channel” panel. See screenshot below. This will add TTX to the sodium ion channel, thus blocking it.



- Now let's see this neuron in action! Select the “Create Simulation” button located under the cell membrane. Select the play button at the bottom of the cell membrane to begin the animation. Take note of what happens in the “Voltage vs. Time” graph.
- Experiment summary explanation: when TTX was added, what happened? Where did TTX bind to and how did this influence the action potential of this neuron (think specifically about the concentrations of sodium and potassium – did they change?) Do you think increasing the intensity of the stimulus would change the result? Why or why not? Feel free to try it!

When TTX was added, it bound to Na⁺ channel and blocked sodium ions entering the cell through here. Sodium was still able to enter through its leak channels, but without the large influx of sodium ions which is needed for an action potential, one never occurred in this experiment. I believe increasing the intensity of the stimulus would have no impact on our results, because with TTX still present the high concentration of Na⁺ that we need cannot enter into cell. My hypothesis was supported by evidence generated by an additional trial I carried out with a 300% increase in the stimulus strength (75μA).

Experiment 7 – Effects of Tetraethylammonium (TEA) on Action Potential Firing

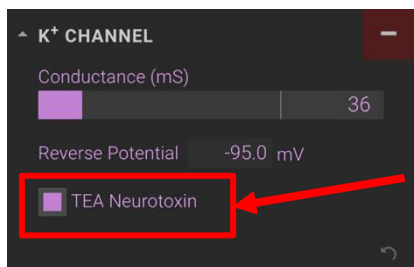
Background: Tetraethylammonium (TEA) is a compound that, like several neurotoxins, was first identified through its damaging effects to the nervous system and shown to have the capacity of inhibiting the function of motor nerves and thus the contraction of the musculature in a manner similar to that of curare. Additionally, through chronic TEA administration, muscular atrophy would be induced. **TEA is a highly specific blocker of potassium channels.**

Steps to complete Experiment 7:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. We want to inject **ONE** stimulus current. Additionally, this current needs to be strong enough for the action potential to reach threshold. As such, change your external current panel to the following:

Current (μA)	Time (ms)
0	10
25	2
0	10
0	2
0	16

4. Next, select the “TEA Neurotoxin” under the “ K^+ Channel” panel. See screenshot below. This will add TEA to the potassium ion channel, thus blocking it.



5. Now let's see this neuron in action! Select the “Create Simulation” button located under the cell membrane. Select the play button at the bottom of the cell membrane to begin the animation. Take note of what happens in the “Voltage vs. Time” graph.
6. Experiment summary explanation: when TEA was added, what happened? Where did TEA bind to and how did this influence the action potential of this neuron (think specifically about the different phases of the AP, such as the rising phase, falling phase, hyperpolarization, duration, and RMP return). EXPLAIN WHY.

When TEA was added, it bound to the K^+ channel and restricted large amount of potassium from leaving the cell (efflux). However, this is not essential to the firing of the action potential, and Na^+ was able to enter the cell like normal here, so an action potential did actually occur in this case. However, hyperpolarization was not able to occur due to the fact that TEA was present. Also, the max voltage was higher (greater action potential) and the time to reach the max voltage (and fire the action potential) was reduced (AP quicker to fire with TEA present).

7. Which Neurotoxin, TTX or TEA, is most effective at inhibiting the action potential? Why?

TTX was more effective than TEA at inhibiting the action potential. This is due to two reasons. The first is that the large influx of Na^+ is needed for an action potential to actually occur, and this cannot happen with TTX present. The second reason is that potassium's 20 times increase in permeability remained a factor in this experiment (and so potassium had a larger impact on membrane potential, in addition to the fact that Na^+ had little effect to begin with due to its low permeability ratio and inability to influx in large concentrations with TTX blocking its channel).